EnGenIR – Engineering Generative transformers with Information Retrieval

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1 Abstract

In facilities engineering, much of an engineer's time is spent reading and understanding technical documentation and standards. Industry standards easily run upwards of hundreds of pages, and dozens of standards may be required to specify a design fully. Confirming the appropriateness and safety of an engineering design takes significant effort and has the potential for catastrophic consequences if done incorrectly. Therefore, our team proposes EnGenIR, a tool that leverages natural language processing (NLP) techniques to answer user design-related queries based on provided technical specifications in the domain of facilities engineering pipeline design. When provided with a user query, this tool uses vector embeddings and information retrieval to identify the most relevant context from the designated specifications. This retrieved context and original user query are passed to a large language model (LLM) in a specifically engineered prompt. The LLM processes this engineered prompt and provides a response to the user. This response is optimized to prioritize accuracy, relevance, and readability and avoid hallucinating information. This tool will provide facilities engineers with valuable assistance in pipeline design and reduction of unnecessary time spent combing through technical standards.

2 Introduction

In today's rapidly evolving world, engineering professionals are confronted with an escalating array of challenges. The relentless advance of technology, coupled with escalating time constraints, necessitates a proficient grasp of increasingly complex systems and design principles. The burgeoning realm of information, compounded by interdisciplinary intricacies, has led to a scenario of 'information overload' where engineers find themselves spending approximately 42% of their time searching for and processing data [Global, 2022]. This inefficient use of time results in significant overhead due to engineering salaries and results in decreased net present value due to delayed project delivery. This manual review process is also error-prone as it requires synthesizing the requirements of multiple standards, each of which may be hundreds of pages long. In some cases, it is a significant task simply to identify all of the relevant applicable standards. Engineering design errors introduce additional costs and delays by requiring additional rework if they are caught before construction. Even more concerning is the risk that errors are not caught at all, as improper design can result in environmental damage and hazards to human life.

Recent advancements in the development of Large Language Models (LLM) have resulted in models with the abilities to understand natural language beyond their training set, generate text that is easily legible and contextually relevant to queries, apply knowledge from the data they were trained on to topics they were not trained on, and reason through problems [Yang et al., 2023]. Our project aims to leverage these LLM capabilities to improve the efficiency and quality of engineering design work by creating an LLM-based assistant that can quickly and accurately answer domain-specific questions posed by engineers.

While LLMs are capable of understanding and making inferences on out-of-domain data, they struggle to reliably provide accurate answers to questions about domains outside of their training data. Updating an LLM to incorporate domain-specific data is computationally expensive and could require changes to the architecture if additional parameters are required [Kaplan et al., 2020]. For added context, let's look at an example. Lambda Labs, a public cloud-based service for training LLMs and other models, estimated that training GPT-3, a previous generation LLM by today's standards, cost over $4.6 million in compute power alone [Labs, 2023]. Additionally, Lambda Labs suggested that training time on a single run would take 355 GPU-years using an NVIDIA V100. Of course, parallelization of GPUs would greatly increase efficiency of this training, but that metric still provides a quantitative understanding of how long it takes to train a comparable LLM from scratch. Now, imagine a small company from a technical industry that wants to leverage an LLM in one of their applications. It is highly probably that this organization would not have the time, money, or in-house expertise to train an original LLM. To take advantage of the power of LLMs, an alternative framework must be considered. It is this business case where our team believes EnGenIR could provide untapped value by leveraging a high-performance, generalized LLM, at significantly lower cost and effort than training a custom model.

In order to overcome the need for training a domain-specific LLM, our project leverages information retrieval paired with a generalized LLM. Information retrieval (commonly abbreviated as IR) is a computer-based process in which a system can retrieve specific information relevant to a provided user query. By retrieving corpus-specific information that is most relevant to the user's query, our team attempts to overcome the aforementioned limitations on domain knowledge. This technique specifically leverages an LLM's ability to make inferences based on provided context, in order to accurately answer a user query. Further details on the utilization of information retrieval are provided in Section 4.1.

Another current limitation of LLMs is that the quality of the response generated is dependent on the quality of the prompt submitted to the LLM. This project additionally and necessarily incorporates prompt engineering to ensure that the LLM is provided clear instructions and the response quality is consistently high. Prompt engineering is a relatively new NLP technique in which the model input is designed to be specific and clear, leading to the generation of a desired output. This technique is discussed further in Section 4.2. This pairing of information retrieval and prompt engineering is how the LLM gains the ability to cite domain-specific requirements from the engineering documents. Additionally, it allows the user notification if the provided query and/or documentation is insufficient to answer the question with high certainty. This combination of user query, relevant context from information retrieval, and a prompt engineering framework to tie it together, is what our team refers to as the 'Power Prompt' (more on this in Section 4).

The scope of this project is focused on the domain of liquid and gas transportation pipeline design, primarily for the oil and gas industry, as part of the larger domain of facilities engineering. Section 3.1 discusses this in greater detail, including the further refinement of the domain into the most common areas of interest. Even so, our goal is to create the framework such that it is flexible enough to generalize to other technical, engineering-based disciplines.

2.1 Objectives

- Build an end-to-end pipeline that can ingest documents and generate vector embeddings, perform information retrieval, and provide relevant and accurate answers to user queries.
- Maximize performance of information retrieval through selection of vector embedding techniques that successfully capture the complex semantics of the facilities engineering domain.
- Leverage generalized state-of-the-art large language models to understand the highly technical facilities engineering domain through the combination of information retrieval and prompt optimization.

2.2 LLM Capabilities and Limitations

Large language models (LLMs), including popular Transformer-based models such as GPT-3 and GPT-4, BERT, and LLaMA [Brown et al., 2020], [OpenAI, 2023], [Devlin et al., 2019], [Touvron et al., 2023] are trained on vast amounts of data collected from many sources: books, articles, various

sites across the internet, etc. (one such example being the Common Crawl data set [Crawl, 2023]). These models capture a wide range of language patterns and knowledge, utilizing word and sentence embeddings as high-dimensional numerical representations of concepts [Mikolov et al., 2013]. The size of the corpus and complexity of language usage mean that these models can answer questions, generate text, and make inferences in a wide variety of domains.

The Transformer-based models consist of many layers of self-attention and feed-forward neural networks. Broadly, they are designed to autoregressively predict text (as tokens) based on their input prompts and any previously predicted text. In their training, they are asked to predict masked words in a sequence (BERT) or the next word in a sequence (GPT). This enables them to learn a model of the probability of any output text given other input text. When run in inference mode, they predict the probability of each of their possible output tokens, and the most likely next token can be chosen. This enables them to perform well in a variety of language-based tasks, and to generate meaningful and relevant novel text across a variety of domains.

The issue with domain-specific performance is that updating LLMs to incorporate new documents can be impractical, due to the cumulative computational costs and the need for significant data science resources for each update [Kaplan et al., 2020]. Moreover, expanding the LLM architecture by incorporating additional parameters becomes necessary to enable memorization of a greater number of facts. While LLMs have shown the ability to learn in-context information from prompts alone [Brown et al., 2020], approaches like Retrieval-Augmented (RetA) models have emerged as promising alternatives [Ram et al., 2023].

RetA models operate by first retrieving relevant context from domain-specific corpora based on user queries. This retrieval is achieved through vector embedding models pre-trained and fine-tuned for semantic retrieval [Neelakantan et al., 2022]. Subsequently, this retrieved context serves as an input to a generalized LLM, enabling the model to provide grounded answers without the need for time-consuming and costly retraining.

Another major drawback of LLMs is their limited knowledge beyond the information they were trained on, resulting in a lack of domain-specific expertise and vocabulary. In addition to their limitations in domain-specific knowledge and biases, LLMs have also been under scrutiny for their potential to generate hallucinated or misleading information [Review, 2022]. This is a key aspect of our project, as engineering designs must rely solely on established facts to ensure safety and reliability.

By combining vector embedding and information retrieval with LLMs, the limitations of the LLMs' domain-specific knowledge and biases can be mitigated. Prompt engineering and the integration of retrieval-augmented information have demonstrated significant improvements in domains such as building information model (BIM) Q&A [Zheng and Fischer, 2023]. Similarly, in the field of biomedical Q&A, the effectiveness of retrieval-augmented models has been highlighted [Neelakantan et al., 2022]. The RetA model demonstrated superior accuracy and relevance, outperforming other high-performance, generalized LLMs in those aspects. Retrieving context from domain-specific corpora allows LLMs to access specialized information and vocabulary, enhancing their understanding and generation capabilities in domain-specific areas. Additionally, retrieval mechanisms help address the concern of generating hallucinated or misleading information by grounding the generated content in verified sources [Neelakantan et al., 2022]. Based on the success of the model pipeline in these fields, our team expected that this process can be successfully applied to the highly technical space of facilities engineering (more specifically pipeline systems design) through our project, EnGenIR.

2.3 Data Science Pipeline

Our project's data science pipeline is shown in Figure 1, and is composed of six parts: Data Acquisition, Pre-Processing, Information Retrieval, Prompt Engineering, LLM Response, and Evaluation & Results. These six components are described in greater detail in the subsequent sections.

Figure 1: Data Science Pipeline

The project team used LangChain [Chase, 2023] to connect the parts of our data science pipeline. LangChain is an open-source architecture for streamlining the development of large language model applications. Consider the analogs of PyTorch or scikit-learn. One could build a neural network or random forest model from scratch, but for production code it is expedient to use existing libraries and functions where available. LangChain facilitates the ability to "connect a language model to other sources of data" and allow it "to interact with its environment" [Chase, 2023]. As its name suggests, this architecture enables the "chaining" of the building blocks necessary to build LLM-based software applications. These building blocks include but are not limited to data ingestion, large language model integration, memory retention, structured sequences known as "chains", and prompt management. In the context of EnGenIR, LangChain is leveraged during our pre-processing, information retrieval, and LLM response phases, in order to increase efficiency and collaboration between each stage. This is explained in greater detail in future sections.

3 Data and Exploration

3.1 Data Acquisition

Our data set is comprised of standards published by the American Society of Mechanical Engineers (ASME) and American Petroleum Institute (API), which are documents 1-3 listed in Table 1. These documents were selected as the key engineering design standards necessary for designing liquid and gas transportation pipelines and also because they were publicly available, avoiding any copyright concerns.

Table 1: Engineering Standards Data

Nr.	**Document Name**	**Pages**	**Size (MB)**
1	ASME B31.4: Pipeline Transportation Systems for Liquids & Slurries [Ame, 2002]	127	10.7
2	ASME B31.8: Gas Transmission & Distribution Piping Systems [Ame, 2004b]	194	17.6
3	API 5L: Line Pipe [Ame, 2004a]	166	8.1

The documents are scanned images of the original documents in PDF format. The loading of these documents requires careful review to ensure accurate conversion to text due to the documents being scanned images and the heterogeneous nature of the document contents. Given the complex nature of the tables, equations, and other contents of the documents, our project focuses solely on understanding the textual content. Furthermore, we have refined this system with the intention of competently addressing queries on a selection of topics related to pipeline design. These topics are outlined in Table 2.

Table 2: Pipeline Design Focus Topics

Material selection	Pressure design principles
Construction methodologies	Assembly protocols
Inspection procedures	Testing strategies
Maintenance procedures	Corrosion control measures

Although this data set is comprised of only three engineering specifications, it represents a substantial amount of text and individual requirements. These three documents include a total word count in excess of 250,000 words, well beyond the token limitation of any mainstream high-performance LLM, including Anthropics's Claude 2 [Anthropic, 2023]. With that said, our team recognizes the possibility that some mainstream LLMs could have included some of this information in their original training set, especially models such as GPT-4, whose full training corpus was never released to the public [OpenAI, 2023]. Preliminary exploration indicates that ChatGPT is capable of answering some simple questions about these specifications, although it does not perform well with more complicated questions or those requiring synthesis of information from multiple sources.

With this in mind, our team also generated a synthetic "company specification" related to pipeline design, which we dubbed EnGenIR 6S. This allowed us to validate the tool's ability to synthesize proprietary information, which can be guaranteed not to have been in the original training set. In the oil and gas industry, it is common for companies to use industry standards while also generating their own supplemental specifications. These specifications can be dozens of pages long, and normally contain clarifications and stricter requirements related to the requirements of the industry specifications. The artificial specification simulated this process by containing supplemental requirements related to various portions of the industry specifications in our corpus. This document was written by members of our team with extensive background in facilities engineering.

Additionally, generalization across other technical domains is a key objective for this project. Therefore, our final model pipeline will allow future adopters of our repository to update the corpus to include any series of technical engineering documents, including proprietary information. The full set of technical documents making up our entire corpus is outlined in Table 3. This set of documents will be fed into the pre-processing step.

Table 3: EnGenIR Corpus

Nr.	**Document Name**	**Pages**	**Size (MB)**
1	ASME B31.4: Pipeline Transportation Systems for Liquids & Slurries [Ame, 2002]	127	10.7
2	ASME B31.8: Gas Transmission & Distribution Piping Systems [Ame, 2004b]	194	17.6
3	API 5L: Line Pipe [Ame, 2004a]	166	8.1
4	EnGenIR 6S (synthetic document)	16	0.25

3.2 Pre-Processing

In order to properly process these engineering design standards, currently in PDF form, they must be loaded and transformed into plain text to allow for further processing to begin. There are a variety of Python-based PDF loaders, such as PyPDF, pdfminer, and pymupdf, to read and process document content. Different PDF loaders offer varying processing speeds and are equipped with their own unique advantages and drawbacks when it comes to working with different types of PDF documents [Miah et al., 2022]. Our team evaluated multiple loaders and selected PyPDF as the most performant for our use case.

The text contained in the PDF documents is tokenized and parsed into chunks of tokens of a specified size, and a specified proportion of overlap between chunks is used in effort to improve the likelihood of capturing relevant content in a chunk. This tokenization is set up to provide the inputs to the vector embedding process, in this case leveraging OpenAI's tiktoken library OpenAI [2021]. One key component of the project was determining the appropriate tools and methods to accurately capture highly technical engineering information. The chunk size should be large enough to contain meaningful information and produce a reasonable number of relevant chunks for a given user query. It must also be small enough for the vector embedding to be able to accurately represent the content and to be workable with LLM input token limits. The chunk overlap should be sufficient to ensure that adjacent chunks are captured by the information retrieval when needed, but not so large as to make the database unwieldy or to overwhelm LLM input token limits. Ideally, each chunk will contain one or a small number of specific requirements. Ultimately, the team settled on a chunk size of 1000 characters, with an overlap of 100 characters between adjacent chunks. This allowed for a reasonable number of chunks to be fed to the LLM within token limits, while generally sorting 1-2 distinct requirements into each chunk.

The data loader allows each chunk to be labeled with both a unique ID and its original source, corresponding to the title of the input document and page from which the content was derived. This metadata is crucial in later stages, as our process requires the LLM to cite each source used in developing an answer. Once the documents have been pre-processed into chunks, these chunks are provided to a vector embedding model for conversion into vector embeddings. This process is discussed in detail in the following section.

3.3 Vector Embeddings

As shown in Figure 2, the document chunks are then converted into embeddings and stored in a vector database. Vector embeddings are commonly used in Natural Language Processing (NLP), as they provide a way to convert text strings into numerical representations which can be understood and manipulated by machine learning models. Each unique token in the vocabulary is assigned a vector of floating point numbers. The distance between any two vectors represents the similarity of their associated tokens, with nearby vectors indicating highly-related tokens. There are many techniques to generate vector embeddings, several of which were explored by our project team. The techniques we have investigated are described in the following sections.

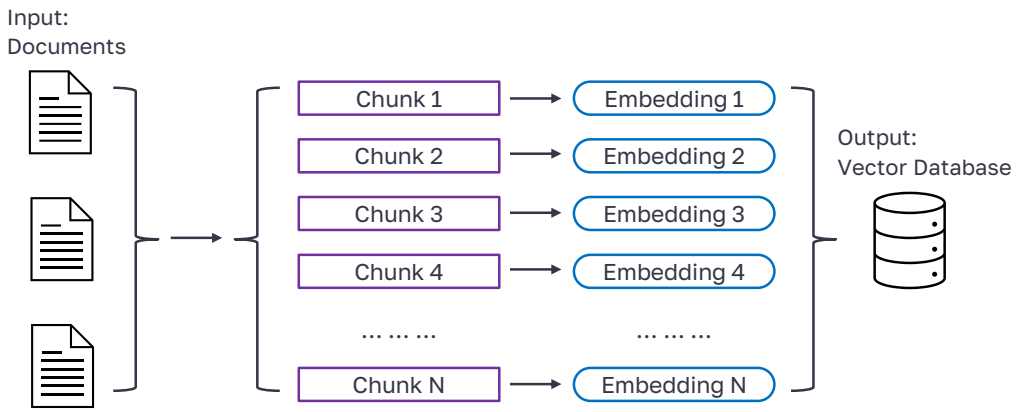


Figure 2: Pre-Processing Overview

To understand the expected performance of a vector embedding model, it is helpful to compare it to a standardized benchmark. In the domain of vector embeddings for NLP tasks, the current gold standard is the Massive Text Embedding Benchmark (MTEB) [Muennighoff et al., 2023]. The MTEB contains 62 datasets that are used for benchmarking text embedding performance across 8 different embedding tasks. Of particular interest to our project is the performance on the retrieval task on the NQ, HotpotQA, FiQA-2018, and TREC-COVID datasets. These datasets test the ability of a query to return a specific article, and the FiQA-2018 and TREC-COVID datasets are from the financial and biomedical domains respectively, which will demonstrate the ability of a generalized embedding model to perform on a domain specific dataset. When comparing models in the following sections, we will reference their performance on the MTEB leaderboard as a baseline for consideration.

3.3.1 Term Frequency-Inverse Document Frequency (TF-IDF)

Term Frequency - Inverse Document Frequency (TF-IDF) is a simple but sometimes effective method for converting documents (or chunks of documents) into vectors [lmu pms, 2017]. In this encoding, words are evaluated based on two metrics: term frequency and (inverse) document frequency. The term frequency (TF) is calculated for each word-document pair, and counts the number of times that the given word appears in the given document. Intuitively, the more frequently a word appears in a document, the more important it is to understanding that document. The document frequency (DF) of a word determines the number of documents in which the word appears. This is then converted into an inverse document frequency:

$$IDF = log(N/DF(w) \tag{1}$$

where N is the number of documents in the corpus and w is the word in question. Intuitively, the more documents which contain a given word, the less information that word conveys about a document. The word "a" appears in most chunks in our corpus, and thus should have very low weight. A rarer word such as "hydrostatic" tells us far more about the chunks in which it appears.

One downside of TF-IDF is that it does not capture significant semantic information, and therefore cannot identify the similarity of documents based on words with similar meanings. It also produces a

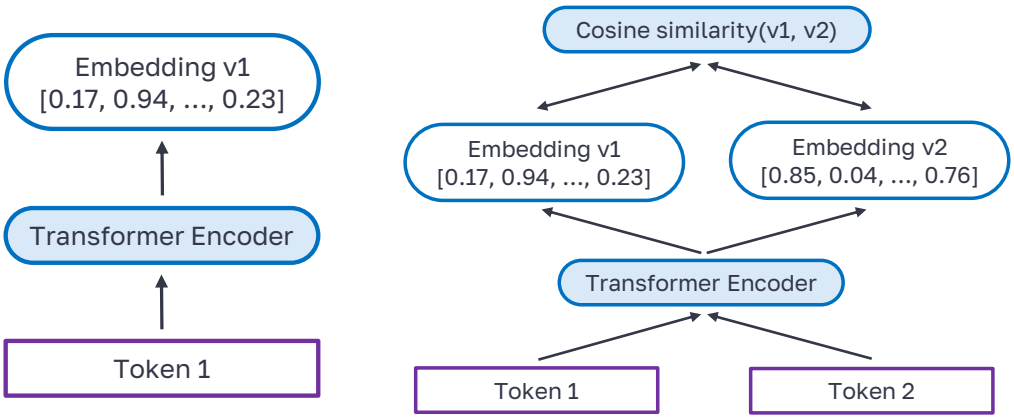
large, sparse embedding matrix by including a dimension for each word in the vocabulary. Lastly, it does not infer any meaning from word order, simple looking at the total counts of each word's appearance.

3.3.2 OpenAI's Text Embedding Ada-002

OpenAI's current state of the art embedding model is text-embedding-ada-002 [Greene et al., 2022]. At the time of this project, text-embedding-ada-002 was ranked sixth overall on the MTEB leaderboard, with competitive performance in the information retrieval tasks. The model also has the advantage of being available through OpenAI APIs, meaning that it can be used to generate embeddings without any GPU requirements for the user. This makes it easier to operationalize the model without having to provide persistent resources for embedding. The downside is that the model is proprietary to OpenAI, so many of its details are not available.

This model trains embeddings using a Transformer encoder on a large corpus of data scraped from the internet. Per [Neelakantan et al., 2022], OpenAI used contrastive learning with in-batch negative sampling, which allows for the use of unlabeled data. Token pairs which appear next to each other in a sample of the text data are considered positive pairs, and negative pairs are randomly sampled from the tokens which do not appear next to each other in the batch. The encoder maps each token to a dense vector embedding of 1536 dimensions (a vector containing 1536 numbers).

The vector embeddings are then trained using cosine similarity of each pair as the loss function. Positive pairs from the data are optimized to maximize their cosine similarity, and negative pairs are optimized to minimize their cosine similarity (see Figure 3). The embeddings are thereby adjusted such that more similar tokens appear closer in the embedding space, and less similar tokens appear farther apart.



(a) Single token embedding with Ada Transformer encoder

(b) Training embeddings using cosine similarity

Figure 3: Mechanism of training Ada embedding model

3.3.3 InstructOR: Instruction-Finetuned Text Embeddings

Another state of the art embedding model is InstructOR (Instruction-based Omnifarious Representations) [Hongjin Su, 2022], introduced in May 2023. At the time of this project, the top-performing version (INSTRUCTOR-XL) was ranked second of all models tested on the Massive Text Embedding Benchmark (MTEB) [Muennighoff et al., 2023], including good performance on information retrieval tasks. InstructOR is based off of an encoder-only version of the T5 Unified Text-to-Text Transformer [Raffel et al., 2020]. InstructOR-XL has 1.5 billion parameters, a maximum input sequence length of 512 tokens, and the resulting dense vector embedding produced by the model has 768 dimensions. It was trained using contrastive loss using a novel dataset of 330 text embedding datasets annotated with human-written task instructions, named MEDI.

The most interesting aspect of InstructOR is the method by which embeddings are fine-tuned for specific tasks. Each input text is provided with a brief instruction regarding its domain and end task,

which is embedded along with the text. The same text is embedded differently based on its end goal. This technique is intended to mitigate the generalization problem, where many embedding models do not perform well on novel tasks or in novel domains. According to the authors, the model was tested on 70 evaluation datasets, 66 of which were fully unseen during training, achieving state of the art performance. This model was of particular interest to the team, as it provides a continuation of the project thesis, namely that prompt engineering of generalized models may be able to outperform fine-tuned models. For information retrieval tasks, the recommendation is actually to use two distinct instructions for the query and the corpus documents to maximize performance.

3.3.4 Microsoft's E5-Large-v2 Text Embeddings

The E5 model developed by Microsoft [Wang et al., 2022] was first released in 2022, and the latest version of the model (E5-Large-v2) was released in 2023 and is composed of 24 layers and 335 million parameters. It is a general-purpose embedding model designed for creating single vector representation of textual input for the purposes of clustering, classification, and information retrieval. At the time of this project, E5-Large-V2 was ranked first overall on the MTEB leaderboard, with strong performance on the information retrieval tasks. Microsoft developed the model by using contrastive pre-training with unlabeled data and fine-tuned using high quality labeled data. The maximum input size to the model is 512 tokens, and the resulting dense vector embedding produced by the model has 1024 dimensions.

3.3.5 Vector Embeddings Applied to EnGenIR

To optimize our model, the team took a staged approach to implementing and evaluating embedding algorithms. The initial implementation centered around two well-established embedding techniques, namely TF-IDF (see Section 3.3.1) and OpenAI's Text Embedding model, Ada-002 (see Section 3.3.2). This allowed us to evaluate the effects of different embedding models and create a viable working product. Once the initial model and project pipeline were set up, we explored additional embedding algorithms to see which could best achieve the project objectives.

TF-IDF is a traditional statistical measure used to reflect how important a word is to a document within a corpus. While simple and computationally efficient, TF-IDF is still an effective way of understanding the significance of terms within large text corpora, making it a reasonable starting point for our project.

On the other hand, Ada-002, an offering from OpenAI, represents a more advanced and modern approach. This model is based on a Transformer architecture, which excels at capturing both the semantic and syntactic nuances of language. Leveraging Ada-002 enabled us to generate embeddings that understand and represent complex engineering terminologies and contexts more effectively.

While TF-IDF and Ada-002 formed the foundation of our embedding strategy, we next explored other models to ensure we utilized the most effective tools. Based on our initial embedding evaluations, discussed in Section 5.1, the Transformer-based architecture provided a significant performance increase. This was unsurprising, given its enhanced ability to capture semantic and syntactic information. To continue this strategy, we implemented and evaluated InstructOR (see Section 3.3.3) and E5 (see Section 3.3.4). This provided two open-source alternatives for comparison to Ada-002, as well as two unique approaches to generating embeddings. The final embedding evaluation results are discussed in Section 5.1.

The chosen embedding model was then applied to convert our document chunks into vector representations. This transformed vector database then served as the foundation for our information retrieval tasks, enabling us to draw upon the power of these sophisticated techniques to deliver precise and meaningful responses.

4 Methods and Models

4.1 Information Retrieval

Figure 4 illustrates the process of information retrieval in a vector database. The first step is conversion of a user query into an embedding, as previously discussed in section 3.3. This embedding then

serves as a key to search the database for the most similar embeddings of chunks from the technical standards.

Specifically, the user prompt is pre-processed and embedded into a vector using the same model which generated the vector database. This newly generated vector is then compared against the vectors in the vector database using cosine similarity [Sv, 2021], which calculates the angle between two vectors (equivalent to their inner product). The objective here is to locate the most similar vectors, which represent the chunks of text that bear the greatest semantic similarity to the query, thereby facilitating effective information retrieval. Going forward, these chunks which are returned by the information retrieval process will be referred to as "context chunks", as they provide context related to the user query.

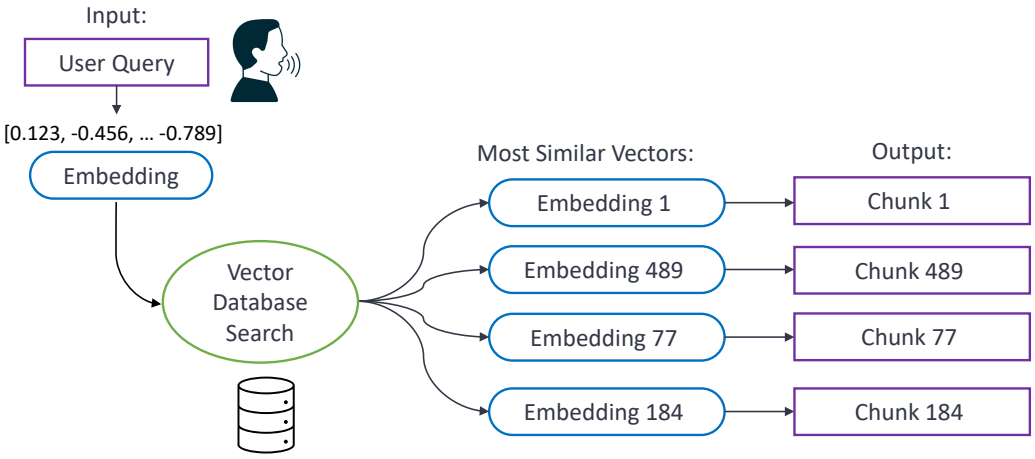


Figure 4: Information retrieval based on user query

Based on this calculated similarity score, the most relevant context chunks from the vector database are then joined together with the user query in order to provide the LLM with sufficient information to answer the user query. The process of joining the query and chunks, along with providing specific instructions to the LLM, involves the NLP domain referred to as prompt engineering. The combination of the user query, relevant context chunks, and prompt engineering will be referred to as the "Power Prompt", as illustrated in Figure 5. Prompt engineering is explained in greater detail immediately below in Section 4.2.

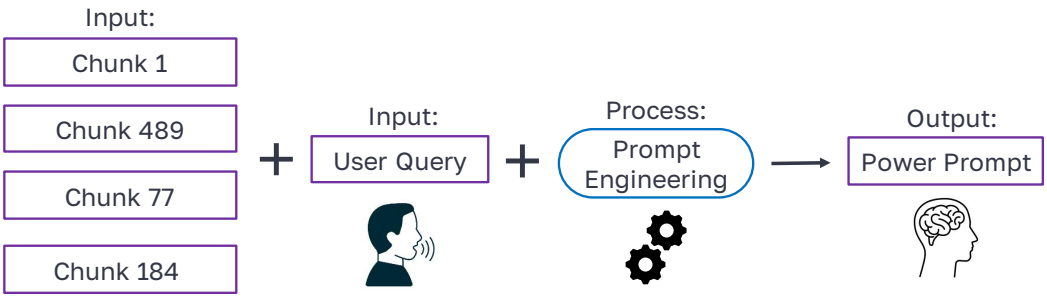


Figure 5: Building a Power Prompt

4.2 Prompt Engineering

Large Language Models have exploded in popularity in recent years, driven largely by the user-friendly interface and general power of ChatGPT. However, LLMs are prone to various issues, including hallucinations, getting "lost" while performing more complex tasks, and a general inability to control the specific format and details of responses. They also are "jacks of all trades, masters of none". A generalized LLM is frequently able to make inferences and perform adequately on many

tasks with either zero-shot or few-shot learning, but is unlikely to have state of the art performance. See 2.2 for additional discussion of these issues.

To improve performance in specific domains, LLMs have been fine-tuned for a variety of tasks, but this is computationally extremely expensive. GPT-3 has 175 billion parameters, and one source [Bastian, 2023] states that GPT-4 has one trillion parameters. The models are trained on gigantic datasets, and even fine-tuning them would require astronomical computational power and data. One alternative to fine-tuning is to use a generalized model, but to condition it for the specific task using prompt engineering.

4.2.1 Evolution of Prompt Engineering

One of the early papers on prompt engineering [Reynolds and McDonell, 2021] compared the technique to the then-favorite few-shot learning as applied to GPT-3. They found that few-shot learning was not teaching the LLM a new task, but rather helping it to identify the appropriate task from its existing space of learned tasks. They further found that careful formatting of prompts could perform this function better, and thereby produce better performance from the model. They took an anthropomorphic approach to prompt engineering, attempting to design prompts which would likely elicit the best performance from a human expert performing the same task. Later papers approach the field with a more scientific point of view.

A later paper [Scao and Rush, 2021] examined the use of prompts in fine-tuning RoBERTa [Liu et al., 2019], an improved pre-trained version of BERT. In their case, the comparison was between traditional fine-tuning vs. fine-tuning using a customized prompt. They found that prompt-based fine-tuning significantly outperformed traditional fine-tuning on 5 out of 6 tasks, and quantified the benefit as the equivalent of training on a much larger data set.

"Pre-train, Prompt, and Predict" [Liu et al., 2021] was one of the earliest to conduct a systematic survey of the discipline of prompt engineering (which they refer to as prompt-based learning) and begin to create a formal grammar. In their formulation, we can consider traditional NLP supervised learning as a discriminative model that takes input text x and predicts output text y based on a model $P(y|x;\theta)$. This model would be trained using a data set of labeled input-output pairs and minimizing the error between model predictions and ground truth outputs. Modern LLMs generally use large amounts of unsupervised data scraped from the Internet. In this case, the training task can be thought of as learning a generative model $P(x;\theta)$ that models the probability of the text itself. This model is then used to generate output by predicting the next most-likely tokens given the input text. In order to improve the performance of the model, prompts can be applied. This can be thought of as applying a "prompting function" to the input text to generate an updated input to the model:

$$x' = f_{prompt}(x)$$

This has been most commonly accomplished by applying prompt templates, which are effectively string wrappers for the input text. The template might have a slot for the original input text, a slot for relevant context, and an additional answer slot to be filled by the model. The authors define prompt engineering as the search for optimal prompts to allow the LLM to solve specific tasks. They further discuss answer engineering, which is the modification of model inputs to constrain the shape or details of the answer we wish to receive.

4.2.2 Prompt Engineering Applied to EnGenIR

Our team considers prompt engineering to be a key aspect of our project pipeline. This technique is the method by which we incorporate retrieved context, ensure that answers are accurate, relevant, and readable, and prevent hallucinations by the model.

One strategy discussed in "Pre-train, Prompt, and Predict" [Liu et al., 2021] is prompt, or in our case query, decomposition. This is a method for breaking a complex task into subtasks. Since LLMs are known to "lose their train of thought" when attempting complex reasoning, our project team used this technique for complex queries to ensure that the model fully answers all sections of a user query and appropriately handles multiple pieces of retrieved context as related to longer, multi-step queries. We implemented this technique in the form of what is considered to be an "LLM agent." An agent is essentially a separate LLM that is specifically primed to handle a task that is distinctly different from the main task at hand. In the case of query decomposition, the LLM agent's only responsibility is to break down the complex, multi-step user query into multiple, concise questions. A simple example

of this technique could be like reducing the following multi-step query "what is the outer and inner diameter?" to two separate queries "what is the outer diameter?" and "what is the inner diameter?"

The concept of breaking prompts into subtasks is further explored by [Wei et al., 2023] in their paper on chain-of-thought prompting. At a high level, this technique consists of requiring the model to work step-by-step through a problem, providing its thought process, critical reasoning, and intermediate answers at each step. They observed that appropriate application of this technique leads to substantial improvements in tasks involving arithmetic, commonsense, and symbolic reasoning. Further, the scale of the improvements is greater for more complex tasks, and for larger models. They achieved state of the art performance on certain tasks. We implemented this technique in the same fashion as query decomposition by using an LLM agent. In continuation of the previous example, the use of this technique would require ingestion of "what is the out diameter?" and "what is the inner diameter" and output critical reasoning steps such as "the outer and inner diameters must be found using engineering specification tables" and "per the engineer specification table X, the outer and inner diameters are..." For this specific example, the LLM agent would be priming the final EnGenIR output to utilize the relevant context consistent with the chain of thought steps–in this case, the engineering specification tables.

The combination of query decomposition and chain-of-thought is a key technique for success in the engineering domain. Our project domain includes complex engineering concepts and complicated reasoning, an area in which chain-of-thought prompting produced substantial improvements. Our model must also be able to explain its reasoning to be useful, and chain-of-thought prompting ensures that the model provides a logical explanation for all of its conclusions. For all complex user queries, our team implemented query decomposition and chain-of-thought reasoning in the development of our Power Prompt.

Table 4 provides a variety of prompt examples, from simplified instructions to more complex techniques like chain of thought reasoning. In this context, prompt refers to any text passed as input to the LLM. These examples include a poorly-written initial user query, an improved user query including retrieved context, and an engineered prompt including retrieved context, a query decomposition prompt, and a chain-of-thought prompt. As previously mentioned, simple queries utilized the shown engineered prompt and proved highly effective. For more complex queries, a multi-step process was used involving the shown query decomposition and chain-of-thought reasoning prompts.

4.3 LLM Response

The use of LLMs allows for the generation of detailed and insightful responses based on the original user query and retrieved context chunks provided in the Power Prompt. This ability plays a vital role in the realm of engineering design validation, where specific, accurate feedback regarding a design's alignment with applicable codes and standards is of the utmost importance.

As illustrated in Figure 6, when the Power Prompt is processed by the LLM, a generated response is produced. For example, if a design adheres to specific codes and standards, the LLM can point out the exact codes, paragraphs, and page numbers in support of its response. Conversely, when a design does not meet certain standards, the LLM can highlight the areas of non-compliance, thus providing valuable insights for further refinement. Additionally, the LLM can even provide feedback back to the user when the query and/or retrieved data is not sufficiently detailed enough and/or does not provide necessary clarity to provide an insightful, and more importantly correct, answer. This an example of how the LLM is "engineered" to prevent hallucinations through the aforementioned technique of prompt engineering.

Additionally, a significant advantage of using an LLM is in its ability to produce human-like responses. This enhances the readability and understanding of the information provided, a feature particularly beneficial for engineers. The clear, actionable feedback delivered by the LLM facilitates an engineer in making informed decisions about their designs. For this reason, our team is leveraging the most easily accessible, high performance, pre-trained model available, which we consider to be GPT-4 OpenAI [2023]. GPT-4 has distinct advantages over GPT-3, specifically in its ability to perform advanced reasoning tasks.

The incorporation of LLMs into the engineering design process can markedly enhance its efficiency and accuracy. By leveraging the LLM's language generation capabilities and information comprehen-

sion, valuable insights can be extracted, compliance issues can be identified, and the decision-making process in engineering design can be supported.

Table 4: Examples of Prompt Engineering

Prompt Type	**Example Prompt**
Simple query	*My design has features A, B, and C. Is that ok?*
Improved query with context	*My oil pipeline has design features A, B, and C. Given context chunks {chunk 1, chunk 2, chunk 3}, does this design comply with all requirements?*
Engineered prompt	*Given the following extracted parts from engineering technical specifications, provide a concise and brief answer to the user query. If the answer cannot be derived from the given information, indicate that more context is needed. Do not attempt to make up an answer.* *User query: {user query}* *Retrieved chunks: {chunk 1, chunk 2, chunk 3}*
Query decomposition	*Given the following user query based on engineering technical specifications and context from engineering technical specifications, assess the need to decompose the query into sub-queries. Examples of queries that may require decomposition include multi-part questions or questions requiring evaluation against multiple specifications. If necessary, perform this decomposition and return the decomposed user query. Otherwise, return 'None'.* *User query: {user query}* *Retrieved chunks: {chunk 1, chunk 2, chunk 3}*
Chain-of-thought reasoning	*Given the following user query based on engineering technical specifications, optional decomposed query, and context from engineering technical specifications, perform chain of thought reasoning to formulate the proper method of evaluation. Pass the critical steps involved in this chain of thought reasoning including justification.* *User query: {user query}* *Decomposed query: {decomp query}* *Retrieved chunks: {chunk 1, chunk 2, chunk 3}*

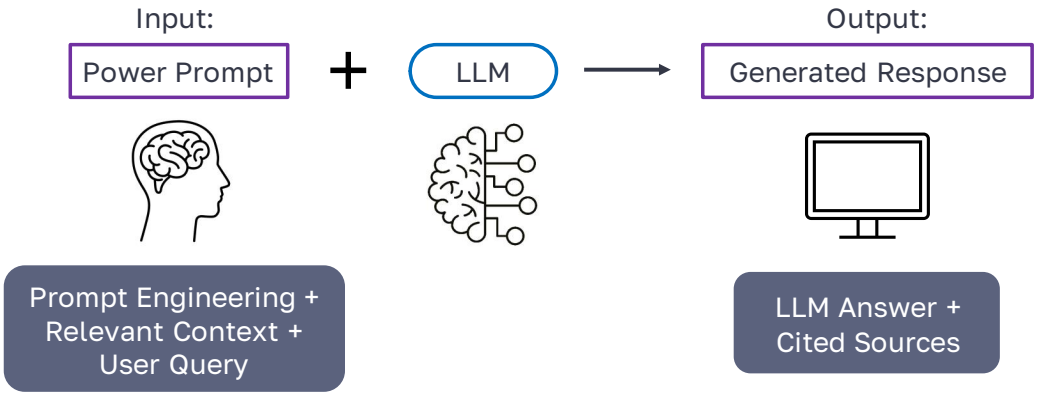


Figure 6: LLM Input & Output

4.4 Full EnGenIR Pipeline

Having discussed the separate components of the data science pipeline in the preceding sections, including Data Acquisition in 3.1, Pre-Processing 3.2, Information Retrieval (IR) in 4.1, Prompt

Engineering in 4.2, and LLM Response in 4.3, we have developed a comprehensive understanding of the individual elements involved in the process. Each of these components forms a critical link in the chain that facilitates the transformation of raw data and user queries into actionable insights.

Now, it's time to assemble these components together and present the full EnGenIR pipeline. This representation will provide a high-level view of the process, demonstrating how each piece connects with and interacts with the others. Refer to Figure 7 for a visual representation of this pipeline. This figure is designed to help us better understand the workflow, tracing the journey of information from the point of data collection to the generation of responses by the LLM. Additionally, Table 7 in Appendix A is available as a walk through of an actual EnGenIR example, showcasing the information utilized during each key step in the pipeline.

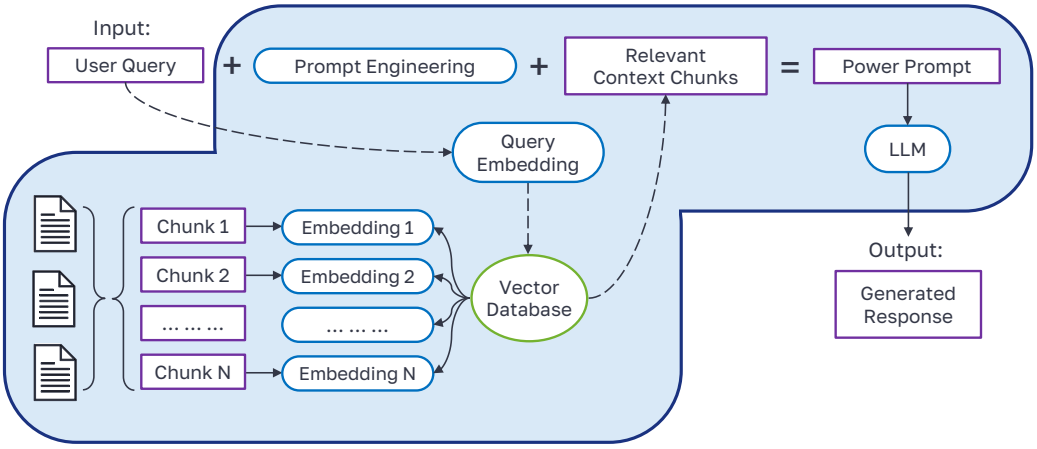


Figure 7: Full EnGenIR Pipeline

The full pipeline commences with the collection of data, chunking, embedding, and uploading it into the vector database. In our case, we have incorporated three Engineering Standards from Section 3.1: ASME B31.4, ASME B31.8, and API 5L. We have also incorporated our synthetic proprietary spec, EnGenIR 6S. Following this, the user enters a query into the model. Based on the user's query, relevant information is retrieved from the vector database. The subsequent step involves generating a Power Prompt, which integrates the initial user query, prompt engineering techniques, and retrieved data. This Power Prompt is then fed to the LLM. The outcome of this comprehensive process is a detailed, insightful, and accurate response to the user's original query.

The full data science pipeline, thus, is a testament to the seamless integration of multiple sophisticated processes. By combining these individual pieces into a unified whole, we are able to effectively convert raw data into meaningful responses, maximizing the potential of LLMs and enhancing the engineering design validation process.

5 Evaluation Framework

An integral part of any project is the evaluation of the model's effectiveness and accuracy. In our project, the primary focus is on evaluating 1) the quality of information retrieval produced by the selected embedding algorithm (its ability to find and return the most relevant context based on a given search query) and 2) the overall output generated by the EnGenIR. Both of these factors have significant implications for the success and reliability of our solution.

5.1 Embedding Techniques Evaluation

Evaluation of embedding techniques focuses on the relevance of the embeddings generated by the embedding technique to user's query. The primary consideration here is whether the embeddings accurately capture the semantic and contextual aspects of the highly technical engineering domain, thus ensuring relevant information retrieval and coherent responses.

To evaluate the quality of the embeddings, we used a set of 216 question-answer (QA) pairs, each pair being based directly on a specific contextual chunk from our corpus. Rather than select these context chunks in chronological order, as they appear in the technical specifications, our team is performing a random selection to ensure there are a variety of topics in the set. This QA set, with representative samples illustrated in Table 5, provides a diverse range of data points, ensuring a thorough and comprehensive evaluation.

Table 5: Examples of Various Domain-Specific Questions

Question	**Topic**
What is the minimum shear area requirements for each heat and the all-heat average of the order based on Charpy Impact Tests for PSL2 on X80?	Inspection Procedures
What is the duration the hydrostatic test pressures for welded pipes that are 20 inches and larger in size?	Testing strategies
Under what conditions, if any, is it permissible to re-use plastic piping that was previously used in natural gas service?	Material selection
What factors should be considered to ensure that all exposed pipe joints can sustain the maximum end force according to the engineering piping codes?	Construction Methodologies
What are the various methods recommended for protecting pipelines from corrosion in the splash zone area, where it is intermittently wet and dry?	Corrosion Control
What is the procedure for preparing pipe that has been used in sour gas service for welding?	Assembly Protocol
What is the maximum gas velocity rate that should be observed in stainless steel piping?	Pressure Design
What documentation/records should be maintained during commissioning of a pipeline system?	Construction Methodologies
How much chromium and copper is allowed in small quantities of certain elements in alloy steels?	Material selection
What is the maximum acceptable depth of grinding to repair external mechanical damage?	Maintenance Procedures
What is the recommended connection point of a service line to a main piping system?	Assembly Protocols

Our embedding techniques evaluation involves using two distinct, but complementary, evaluation metrics: Recall @ k and Normalised Discounted Cumulative Gain (nDCG@k). These evaluation metrics provide us with a quantifiable measure of the quality and relevance of the embeddings discussed in Section 3.3 Vector Embeddings.

Recall@k: The Recall@k metric provides an assessment of whether the correct chunk from the QA pair is actually retrieved in the top-k results [Carnevali]. In the context of our evaluation, it essentially gauges whether our embedding technique successfully places the most relevant QA pairs within the top k retrieved pairs. This technique does not take the order of the retrieved pairs into account and is only concerned that the relevant pairs are captured in the top k results. This is the primary evaluation metric as k number of chunks will be passed to the prompt engineering phase, and the order of the chunks is irrelevant as long as the relevant chunks are captured. Recall@k can be calculated using Equation 2.

$$Recall@k = \frac{\textit{Number of relevant items in the top } k \textit{ recommendations}}{\textit{Total number of relevant items}} \tag{2}$$

Normalized Discounted Cumulative Gain (nDCG@k): Unlike Recall@k, nDCG@k takes into account not just the presence of relevant chunk, but also its position in the resultant list Carnevali. nDCG@k is a position-sensitive metric, which assigns higher value to relevant items appearing at higher ranks (i.e., towards the top of the list). This is particularly important in contexts where the ranking of results matters, as is often the case in information retrieval tasks. The "normalized" part of nDCG@k means that the results are scaled between 0 (worst) and 1 (best) so that they can be

compared across different queries and contexts. This was used to discriminate between embeddings techniques that result in similar Recall@k values, as the order that the chunks are retrieved could indicate the ability of a model to better generalize outside of our test set.

nDCG@k is calculated using Equations 3 and 4, where *DCG@k* is the discounted cumulative gain, *IDCG@k* is the ideal discounted cumulative gain which is calculated the same way as *DCG@k* except with all items perfectly ranked, and rel_i is the relevance score of the item at position *i*.

$$NDCG@k = \frac{DCG@k}{IDCG@k} \tag{3}$$

$$DCG@k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)} \tag{4}$$

nDCG@k and Recall@k computations are implemented in EnGenIR by using the industry standard pytrec_eval Python packaged [Van Gysel and de Rijke, 2018]. By using these techniques, we hope to ensure that our embeddings not only capture the semantic and contextual intricacies of the engineering domain but also facilitate efficient and meaningful information retrieval.

5.2 EnGenIR Output Evaluation

The second part of our evaluation was focused on the overall output generated by the system. For this evaluation, we used a set of 69 question-answer pairs curated by industry experts with 25+ years experience. These experts provided high-quality, relevant questions and correct answers, giving us a reliable gold standard against which we evaluated the system.

Upon conducting an exhaustive review of existing literature, our team chose to adopt an evaluation framework in line with the RetA model, as detailed in [Ram et al., 2023]. We selected this framework based on its close alignment with our project's objectives and to maintain standardization and consistency in our evaluation methodology. In terms of evaluation criteria, we assessed the LLM's generated output based on a 3-point scale, measuring accuracy, relevance, and readability. Detailed descriptions of these criteria can be found in Table 6.

Accuracy: This assesses the factual correctness of answers as related to the cited source and technical specifications.

Relevance: This focuses on how well the model understands the question and delivers an appropriate answer in the domain of engineering.

Readability: This is based on factors including but not limited to clarity, coherence, grammatical correctness, and use of domain-specific language.

Table 6: Evaluation Metric Descriptions [Ram et al., 2023]

Metric	Score 1	Score 2	Score 3
Accuracy	Mostly inaccurate or misleading content	A mix of accurate and inaccurate content	Factually accurate and reliable content
Relevance	Mostly irrelevant content	Partially relevant content	Highly relevant and on-point content
Readability	Difficult to read, unclear or convoluted language	Moderately readable, with some passages	Easy to read, clear, and concise language

These ratings were utilized during the model's performance optimization process and for the documentation of the final results. We believe this methodology ensured consistency in the evaluation process.

It's imperative to note that the pool of 69 questions was divided into categories of "Easy" and "Difficult," with 50 questions designated as easy and 19 as difficult. Subsequently, these question categories were evenly divided into training and testing subsets, accounting for 70% and 30% respectively. This partitioning strategy ensured that both the training and testing sets contained an

equivalent proportion of easy and difficult questions. The training set was instrumental in refining prompt engineering techniques, encompassing prompt decomposition and chain-of-thought reasoning.

Finally, the test set, consisting of 23 questions, was employed to perform a blind evaluation of the answers furnished by EnGenIR, Claude 2, and GPT-4. To truly understand the performance of our model, we felt it was important to benchmark against the zero-shot performance of state of the art generalized LLMs. In the current landscape, GPT-4 and Claude 2 are among the top performing models.

6 Results

6.1 Embedding Benchmarking Results

As discussed in section 5.1, our team has evaluated the effectiveness of multiple embedding techniques in order to maximize information retrieval for the relatively complex contents of facilities engineering specifications.

As seen in Figure 8 which displays the Recall@k scores at various values values of k, our benchmarking results show that Open AI's text-embedding-ada-002 and Microsoft's E5_Large_v2 were the top performing models, with E5_Large_v2 slightly outperforming Ada-002 for k values 1-2 and 7-10. The E5_Large_v2 model achieved a recall@k of 75.9% at k=5 and 84.0% at k=10, compared with Ada-002 delivering 76.4% and 82.1% respectively.

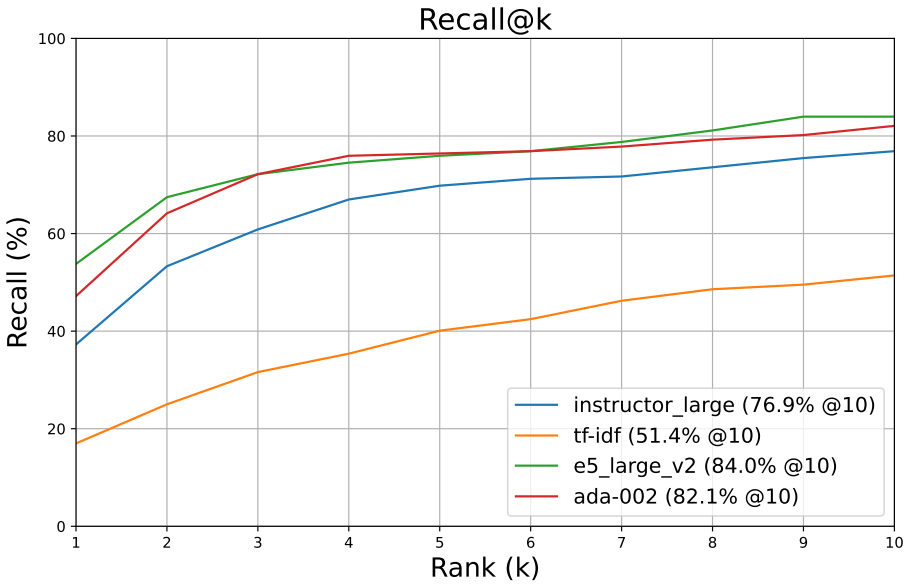


Figure 8: Embeddings Benchmarking Results using Recall@k

Additionally, as seen in Figure 9 which displays Normalized Discounted Cumulative Gain (nDCG) scores at various values of k, the relative performance gap between all models remains the same when scoring based on this alternate metric. Ada-002 scores 63.6% for k=5 and 65.5% for k=10, while E5_Large_v2 scores 66.3% and 68.9% for the same respective k.

The minor differences in model performance between the Ada-002 and E5_Large_v2 afforded the team flexibility in choosing a model based on where the model would be run. Ada-002 was chosen due to the ability to run the model with API calls to Open AI's servers, which allowed for heavy computation to be performed off of the device and makes deployment of EnGenIR more practical for users. This added flexibility comes at a financial cost of paying for API calls which would not exist for the E5_Large_v2 model. Future users could select either Ada-002 or E5_Large_v2 to suit their own needs and preferences, knowing that there would be no significant change in performance.

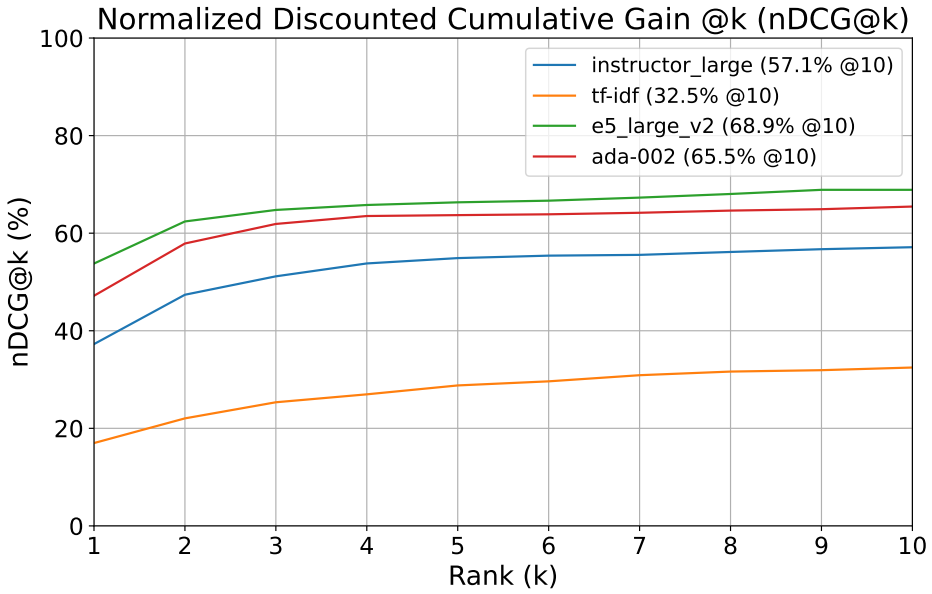


Figure 9: Embeddings Benchmarking Results using nDCG@k

6.2 EnGenIR Output Results

As discussed in section 5.2, our team evaluated the the final outputs of EnGenIR based on a 3-point scale for 3 metrics: accuracy, readability, and relevance. The final results, presented in Tables 8, 9, and 10 of Appendix B, display the average of the team's ratings based on 23 questions from Appendix C, curated by the professional Facilities Engineers within our team.

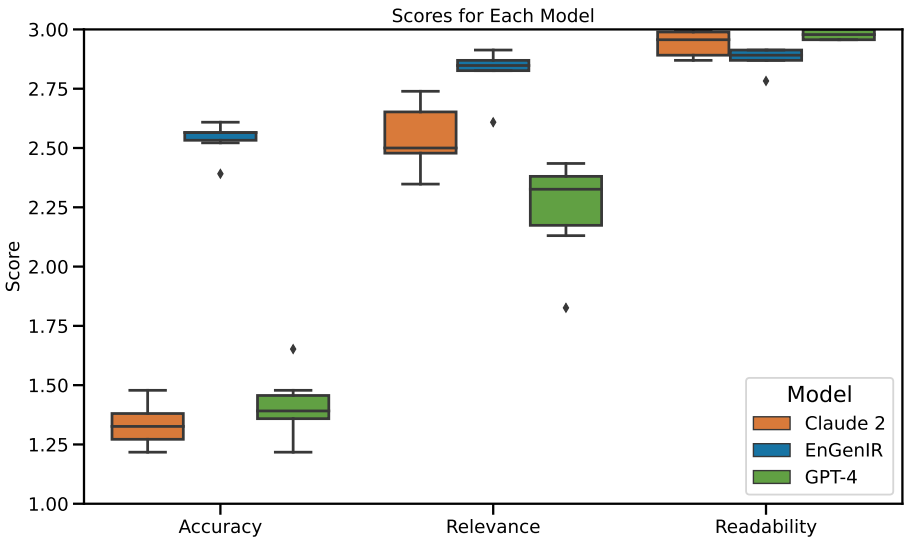


Figure 10: Final Model Evaluation Scores

In addition to ranking the absolute scores for EnGenIR outputs, we also compared its performance with two state of the art generalized LLMs: Claude 2 and GPT-4. We expected that a generalized LLM not conditioned on a specific corpus would exhibit inferior performance compared to EnGenIR,

although advanced models like GPT-4 would likely still demonstrate high readability. The primary advantage of EnGenIR lay in its capability to confine responses within the scope of provided context, its proficiency in utilizing information beyond the training data of publicly available general LLMs, and its ability to correctly synthesize multiple data sources to answer more complex queries.

In order to retrieve answers from EnGenIR, we employed the Power Prompt technique for all questions. During the retrieval process, EnGenIR provided a "More context is needed" response for 3 out of 23 test questions, as it had not found a relevant piece of information within the retrieved text chunks. To address these cases, we applied Prompt Decomposition and Chain-of-thought reasoning techniques, enabling EnGenIR to generate answers for these complex questions.

When comparing EnGenIR to generalized LLMs—Claude 2 and GPT-4—in our study, we discovered that while generalized LLMs demonstrated high Readability scores, EnGenIR surpassed them in terms of Relevancy and was far superior in terms of Accuracy as seen in Figure 10.

7 Discussion

Although overall considered a success, there are several avenues by which the current model could be expanded. One key improvement would be the further development of the pre-processing technique to allow for inclusion of tabular data and equations. Many engineering tasks involve value look-ups in tables or performing calculations, and allowing the LLM to support these tasks would add significant value. Additional work in pre-processing could include hyperparameter tuning to select the ideal chunk size and overlap for a given document or domain. A truly interesting method would be semantic chunking of documents, whereby the document would be divided up into individual requirements, rather than chunks of a fixed size. Our team also examined a number of vector embedding algorithms, but there are many more and improvements are constantly being made. Further efforts could examine additional algorithms and attempt to optimize further for a given domain.

The use of information retrieval to augment LLM responses is not new. However, our project is the first of which we are aware that applies the technique to the domain of Facilities Engineering. Given the rapid pace of change, we believe this project is also among the first to apply retrieval augmentation to GPT-4 and to compare results to GPT-4 and Claude 2. We are unaware of any documented attempts to measure the accuracy of recommendations made by Facilities Engineers, so we unfortunately cannot compare the performance of EnGenIR to that of a human engineer. Nonetheless, we believe that the model performance indicates that it is a worthwhile tool to aid human engineers in their daily tasks. We further believe that the framework we have developed is flexible enough to be scale with additional technical documentation and across other technical domains. The development and implementation of EnGenIR has been a rewarding experience. Overall, our team is satisfied with the model's performance and look forward to the potential this innovative application brings in the future.

A Actual Example of EnGenIR

Table 7: Actual Example of EnGenIR for k=10

User Query:
user query = "We are going to use 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig. The plastic piping wall thickness is 0.1". Can you assess the design?"
Relevant Context Chunks:

Continued on next page

"itations (a) The value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. (b) Reinforced thermosetting plastic pipe and fittings shall not be used where operating temperatures will be below -200P or above 150°F. (c) The value of t for reinforced thermosetting plastic pipe shall not be less than that shown in Table 842.33(c). 842.34 Design Pressure of Plastic Fittings. The maxi¬ mum pressure rating for fittings shall be the same value as the maximum design pressure of the corresponding pipe size and wall thickness as indicated in the refer¬ enced standard for the fittings and as determined in paras. 842.31 and 842.32. The manufacturer should be 42 Table 842.33(c) Diameter and WaU Thickness for Reinforced Thermosetting Plastic Pipe Minimum Nomina! Pipe Outside Diameter, Thickness, Size in. in. 2 2.375 0.060 3 3.500 0.060"

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"peratures above 150°F. The duration of the test of thermoplastic piping above 100°F, however, shall not exceed 96 hr. (c) Sufficient time for joints to "set" properly must be allowed before the test is initiated. (d) Plastic pipelines and mains shall be tested at a pressure not less than 1.5 times the maximum operating pressure or 50 pSig, whichever is greater, except that (1) the test pressure for reinforced thermosetting plastic piping shall not exceed 3.0 times the design pres¬ sure of the pipe (2) the test pressure for thermoplastic piping shall not exceed 3.0 times the design pressure of the pipe at temperatures up to and including 1000P or 2.0 times the design pressure at temperatures exceeding 1000P 46 842.44-843.11 (e) Gas, air, or water may be used as the test medium. 842.53 Safety During Tests. All testing after con¬ struction shall be done with due regard for the safety of employees and the public during the test. 842.6 Copper Mains"

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"nesses. See Appendix of ANSI A21.52. nominal wall thickness for a given design pressure (sub¬ ject to the limitations in para. 842.32) shall be determined by the following formula: where o p s I P = 2S (D - t) x 0.32 specified outside diameter, in. design pressure, psig for thermoplastic pipe and tubing, long-term hydrostatic strength determined in accordance with the listed specification at a temperature equal to 73°F, 100°F, 120°F, or 140oP; for rein¬ forced thermosetting plastic pipe, use 11,000 psi 41 t = specified wall thickness, in. NOTE: Long-term hydrostatic strength at 73°F for the plastic mate¬ rials whose specifications are incorporated by reference herein are given in Appendix D. 842.32 Thermoplastic Design Limitations (a) The design pressure shall not exceed 100 psig. (b) Thermoplastic pipe, tubing, and fittings shall not be used where the operating temperatures of the materi¬ als will be (1) below -20oP, or"

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...

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"greater depth, or indicating the presence of the line with additional markers. 402.2 Pressure-Temperature Ratings for Piping Components 402.2.1 Components Having Specific Ratings. Within the metal temperature limits of -20°F (-30°C) to 250°F (120°C), pressure ratings for components shall 9 401.6.1-402.3.1 conform to those stated for 1 OO°F (40°C) in material standards listed in Table 423.1. The nonmetallic trim, packing, seals, and gaskets shall be made of materials which are not injuriously affected by the fluid in the piping system and shall be capable of withstanding the pressures and temperatures to which they will be sub¬ jected in service. Low temperatures due to pressure reduction situations, such as blow downs and other events, shall be considered when designing carbon dioxide pipelines. 402.2.2 Ratings -Component Not Having Spe¬ cific Ratings. Piping components not having established pressure ratings may be qualified for use as specified"

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Chain of Thought:

Chain of Thought Reasoning:
1. The first step is to assess whether the use of 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location is acceptable according to the provided technical specifications. From the context, it is clear that plastic piping may be used within the limitations prescribed in the Code for Class 4 locations. Therefore, the use of plastic piping in this scenario is acceptable.
2. The second step is to evaluate whether the operating temperature of 40°F is within the acceptable range for the use of plastic piping in a gas transmission system. According to the context, reinforced thermosetting plastic pipe and fittings should not be used where operating temperatures will be below -200°F or above 150°F. Since the operating temperature of 40°F falls within this range, it is acceptable.
3. The third step is to determine whether the pressure not exceeding 100 psig is within the acceptable range for the use of plastic piping in a gas transmission system. The context states that the value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. Therefore, the pressure of 100 psig is within the acceptable range.
4. The final step is to assess whether the plastic piping wall thickness of 0.1" is acceptable according to the provided technical specifications. The context provides a table that shows the minimum wall thickness for reinforced thermosetting plastic pipe of different sizes. For a 6" pipe, the minimum wall thickness is 0.100". Therefore, the wall thickness of 0.1" is acceptable.
In conclusion, based on the provided technical specifications, the design of using 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig, with a plastic piping wall thickness of 0.1", is acceptable.'

Intermediate Steps:

'chunk id': 285.0 : "(a) The value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. (b) Reinforced thermosetting plastic pipe and fittings shall not be used where operating temperatures will be below -200P or above 150°F. (c) The value of t for reinforced thermosetting plastic pipe shall not be less than that shown in Table 842.33(c). 842.34 Design Pressure of Plastic Fittings. The maxixadmum pressure rating for fittings shall be the same value as the maximum design pressure of the corresponding pipe size and wall thickness as indicated in the referxadenced standard for the fittings and as determined in paras. 842.31 and 842.32. The manufacturer should be 42 Table 842.33(c) Diameter and WaU Thickness for Reinforced Thermosetting Plastic Pipe Minimum Nomina! Pipe Outside Diameter, Thickness, Size in. in. 2 2.375 0.060 3 3.500 0.060",
'chunk id': 308.0 : No relevant text found

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Table 7–Continued from previous page

'chunk id': 281.0 : "The design pressure shall not exceed 100 psig. Thermoplastic pipe, tubing, and fittings shall not be used where the operating temperatures of the materials will be below -20oP."
...
'chunk id': 103.0 : No relevant text found

Power Prompt:

User Query: "We are going to use 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig. The plastic piping wall thickness is 0.1". Can you assess the design?"

Chain of Thought: 1. The first step is to assess whether the use of 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location is acceptable according to the provided technical specifications. From the context, it is clear that plastic piping may be used within the limitations prescribed in the Code for Class 4 locations. Therefore, the use of plastic piping in this scenario is acceptable.
2. The second step is to evaluate whether the operating temperature of 40°F is within the acceptable range for the use of plastic piping in a gas transmission system. According to the context, reinforced thermosetting plastic pipe and fittings should not be used where operating temperatures will be below -200°F or above 150°F. Since the operating temperature of 40°F falls within this range, it is acceptable.
3. The third step is to determine whether the pressure not exceeding 100 psig is within the acceptable range for the use of plastic piping in a gas transmission system. The context states that the value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. Therefore, the pressure of 100 psig is within the acceptable range.
4. The final step is to assess whether the plastic piping wall thickness of 0.1" is acceptable according to the provided technical specifications. The context provides a table that shows the minimum wall thickness for reinforced thermosetting plastic pipe of different sizes. For a 6" pipe, the minimum wall thickness is 0.100". Therefore, the wall thickness of 0.1" is acceptable.
In conclusion, based on the provided technical specifications, the design of using 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig, with a plastic piping wall thickness of 0.1", is acceptable.

Relevant Context:

'chunk id': 285.0 : "(a) The value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. (b) Reinforced thermosetting plastic pipe and fittings shall not be used where operating temperatures will be below -200P or above 150°F. (c) The value of t for reinforced thermosetting plastic pipe shall not be less than that shown in Table 842.33(c). 842.34 Design Pressure of Plastic Fittings. The maxixadmum pressure rating for fittings shall be the same value as the maximum design pressure of the corresponding pipe size and wall thickness as indicated in the referxadenced standard for the fittings and as determined in paras. 842.31 and 842.32. The manufacturer should be 42 Table 842.33(c) Diameter and WaU Thickness for Reinforced Thermosetting Plastic Pipe Minimum Nomina! Pipe Outside Diameter, Thickness, Size in. in. 2 2.375 0.060 3 3.500 0.060",
'chunk id': 308.0 : No relevant text found
'chunk id': 281.0 : "The design pressure shall not exceed 100 psig. Thermoplastic pipe, tubing, and fittings shall not be used where the operating temperatures of the materials will be below -20oP."
...
'chunk id': 103.0 : No relevant text found

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Table 7–Continued from previous page

Final Prompt: Given the following user query, chain of thought reasoning, and context from engineering technical specifications, provide a brief and concise answer to the user query. Refer back to the technical specifications for your response ("SOURCES").
If the answer cannot be derived from the given information, indicate that more context is needed. Do not attempt to make up an answer.
Always provide a "SOURCES" section in your response, highlighting which part of the technical specifications your answer is based on.
Final Answer:

Generated Response:

"Yes, the design of using 6” plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig, with a plastic piping wall thickness of 0.1”, is acceptable according to the provided technical specifications."

SOURCES:
- The value of P for reinforced thermosetting plastic mains and service lines in distribution systems in all class locations and for other applications in Class 3 and 4 locations shall not exceed 100 psig. (inputs/asme.b31.8.2003.pdf)
- Reinforced thermosetting plastic pipe and fittings shall not be used where operating temperatures will be below -200°F or above 150°F. (inputs/asme.b31.8.2003.pdf)
- The value of t for reinforced thermosetting plastic pipe shall not be less than that shown in Table 842.33(c). (inputs/asme.b31.8.2003.pdf)

B Final EnGenIR Outputs Assessment

Table 8: Accuracy Ratings of Output based on Test Set

Number	Claude 2	EnGenIR	GPT-4
9	1.17	2.83	1.33
69	1.00	1.00	1.17
63	1.00	3.00	1.00
61	1.50	2.50	1.17
60	2.67	3.00	2.00
59	3.00	3.00	2.00
58	1.50	3.00	2.50
51	1.50	3.00	2.00
47	1.00	3.00	1.00
37	1.00	3.00	1.00
35	1.00	2.00	1.00
31	1.00	3.00	1.00
27	1.17	1.50	1.17
26	1.00	1.83	1.67
24	1.00	3.00	1.33
23	1.00	3.00	1.00
21	1.00	1.00	1.00
16	1.67	2.00	1.00
15	1.00	2.00	1.83
74	1.00	2.83	1.00
73	1.00	3.00	1.83
72	2.00	3.00	2.00
71	1.50	2.83	1.33
Average	1.33	2.54	1.41

Table 9: Relevance Output Ratings based on Test Set

Number	Claude 2	EnGenIR	GPT-4
9	3.00	3.00	2.00
69	2.33	1.17	2.00
63	2.83	3.00	2.83
61	2.67	3.00	2.00
60	3.00	3.00	2.67
59	3.00	3.00	3.00
58	2.67	2.67	2.67
51	2.33	2.83	2.33
47	1.67	2.83	1.17
37	2.67	2.83	1.33
35	2.17	2.67	1.17
31	2.00	3.00	1.67
27	1.83	2.33	1.83
26	1.83	2.83	2.17
24	2.00	3.00	2.00
23	2.83	3.00	2.17
21	3.00	2.83	1.50
16	2.67	2.83	2.83
15	2.50	3.00	2.50
74	2.67	3.00	3.00
73	3.00	3.00	3.00
72	3.00	3.00	2.67
71	2.83	3.00	2.67
Average	2.54	2.82	2.24

Table 10: Readability Output Ratings based on Test Set

Number	Claude 2	EnGenIR	GPT-4
9	3.00	3.00	3.00
69	3.00	3.00	3.00
63	3.00	3.00	3.00
61	2.67	2.50	2.67
60	3.00	3.00	3.00
59	3.00	3.00	3.00
58	2.50	2.83	3.00
51	3.00	2.67	3.00
47	3.00	3.00	2.83
37	2.83	3.00	3.00
35	2.83	2.83	3.00
31	3.00	3.00	3.00
27	2.83	1.33	3.00
26	3.00	3.00	3.00
24	2.83	3.00	3.00
23	3.00	3.00	3.00
21	3.00	3.00	3.00
16	3.00	3.00	3.00
15	3.00	3.00	3.00
74	3.00	3.00	3.00
73	3.00	3.00	3.00
72	3.00	3.00	3.00
71	3.00	3.00	3.00
Average	2.94	2.88	2.98

C Questions and Verified Answers Test Set

Table 11: Questions and Verified Answers in Test Set

Number	Question	Verified answer
9	I am in the middle of constructing a gas transmission system and have questions about the flange joints. We are trying to bolt steel flanges and have only ASTM A307 Grade B bolts available. Can I use them?	Yes, but only for ANSI B16.5 Class 150 and 300 flanges at temperatures between -20 F and 450°F. ASME B31.8
69	What is the maximum lapped flange thickness for welding neck flanges?	ASME B31.4: The maximum lapped flange thickness for welding neck flanges is 1/2 in. (13 mm)
63	What is the minimum duration for hydrostatic tests to prove strength of pipelines and mains to operate at hoop stresses of 30% or more of the specified minimum yield strength of the pipe?	ASME B31.8: The minimum duration for hydrostatic tests to prove strength of pipelines and mains to operate at hoop stresses of 30% or more of the specified minimum yield strength of the pipe is 2 hours.
61	What are the techniques that can be used to provide additional protection from third-party damage to a buried main or pipeline crossing within or parallel to the right-of-way of a road or railroad?	ASME B31.8: Techniques that can be used to provide additional protection from third-party damage to a buried main or pipeline crossing within or parallel to the right-of-way of a road or railroad include: (1) a concrete or steel barrier placed above the pipe, (2) a concrete slab placed vertically adjacent to the pipe on each side and extended above the top of pipe elevation, (3) damage-resistant coating material, such as concrete, (4) extra depth of cover additional to that required in para. 841.142, and (5) buried high-visibility warning tape placed parallel to and above the pipe.
60	What measures should be taken to mitigate the effects of AC voltages on personnel safety during pipeline construction and operation?	ASME B31.8: Measures to mitigate the effects of AC voltages on personnel safety during pipeline construction and operation include bonding, shielding, or grounding techniques.
59	What is the minimum cover requirement for service lines in private property?	ASME B31.8: The minimum cover requirement for service lines in private property is 12 inches.
58	What special considerations are necessary for the design and maintenance of offshore pipeline facilities that may carry natural gas containing corrosive elements?	ASME B31.8: Special consideration for the control of internal corrosion is necessary for the design and maintenance of offshore pipeline facilities that may carry natural gas containing carbon dioxide, chlorides, hydrogen sulfide, organic acids, solids or precipitates, sulfur-bearing compounds, oxygen, or free water.

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Table 11–Continued from previous page

Number	Question	Verified answer
51	What are the suitable types of protective devices to prevent overpressuring of high-pressure distribution systems?	ASME B31.8: Suitable types of protective devices to prevent overpressuring of high-pressure distribution systems include relief valves, weight-loaded relief valves, a monitoring regulator installed in series with the primary pressure regulator, a series regulator installed upstream from the primary regulator and set to limit the pressure on the inlet of the primary regulator continuously to the maximum allowable operating pressure of the distribution system or less, and an automatic shutoff device installed in series with the primary regulator.
47	What areas should be inspected for atmospheric corrosion?	ASME B31.8: Areas that should be inspected for atmospheric corrosion include flanges, flange bolts, areas under pipe straps, areas where pipe is in contact with supports, and other places where moisture collects
37	Inspection revealed a dent on an NPS 16 steel pipe, which does not exceed a maximum depth of 1% of the nominal pipe diameter, and the pipeline is intended to operate at hoop stress levels of 40% or more of the specified minimum yield strength. What shall I do?	ASME B31.8: if less than 2% - OK
35	We are designing the gas transmission system within a class 4 location, where no fewer than 75% of welds shall be examined. Are there any exceptions to this requirement?	ASME B31.8: When pipe size is less than NPS 6, or when the construction project involves such a limited number of welds that nondestructive inspection would be impractical, and the pipe is intended to operate at hoop stress levels of 40% or less of the specified minimum yield strength, then provisions (b) and (c) above are not mandatory, provided the welds are inspected visually and approved by a qualified welding inspector.
31	Under what conditions, if any, is it permissible to re-use plastic piping that was previously used in natural gas service?	ASME B31.8: Used plastic pipe and tubing of known speCifications and dimensions that has been used in natural gas service only may be reused, provided that (a) it meets the requirements of ASTM D 2513 for new thermoplastic pipe or tubing, or ASTM D 2517 for new thermosetting pipe (b) a careful inspection indicates that it is free of visible defects (c) it is installed and tested in accordance with the requirements of this Code for new pipe
27	What is the procedure for preparing pipe that has been used in sour gas service for welding?	ASME B31.8: Pipe that has been used in sour gas service shall be heated for at least 20 min at 400°F (204°C) or higher to drive off any hydrogen in the metal. Heating shall be done just prior to welding. This heating should be in addition to and immediately preceding any preheating specified in the welding procedure for new pipe.

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Table 11–Continued from previous page

Number	Question	Verified answer
26	What is the approved method for permanently repairing small corroded areas on steel pipelines operating at hoop stress levels at or above 30% of the specified minimum yield strength? How do pressure and flow rate affect the likelihood of burn-through in this process?	ASME B31.8: 851.44: 851.44 Permanent Field Repair of Leaks and NonLeaking Corroded Areas (a) If feasible, the pipeline shall be taken out of service and repaired by cutting out a cylindrical piece of pipe and replacing with pipe of equal or greater design strength...
24	We are going to use 6" plastic piping as a temporary solution for the gas transmission system within a Class 4 location, where the operating temperature is 40°F, and the pressure does not exceed 100 psig. The plastic piping wall thickness is 0.1". Can you assess the design?	According to ASME B31.8, everything looks good for this design
23	What is the duration the hydrostatic test pressures for welded pipes that are 20 inches and larger in size?	API 5L: Test pressures for welded pipe in sizes 20 and larger shall be held for not less than 10 seconds.
21	What is the minimum shear area requirements for each heat and the all-heat average of the order based on Charpy Impact Tests for PSL2 on X80?	API 5L: For X80 only, the required minimum shear area shall be either 40% for each heat and 70% for the all-heat average of the order based on the Charpy test, or 40% for each heat and 60% for the all-heat average based on the drop-weight tear test. The drop-weight tear test option only applies for welded pipe in sizes 20 or larger. If the all-heat average of the order does not meet the required percentage of shear area, the manufacturer shall be responsible for the replacement of heats as necessary to bring the average up to the required level.
16	A pipeline constructed with API 5L X70 material will cross several fault lines in a seismically active region. What type of stress and strain tests should be performed to assess the pipeline's resilience to potential seismic activities?	(B31.4) When an offshore pipeline is to be laid across a known fault zone or in an earthquake-prone area, consideration shall be given to the need for flexibility in the pipeline system and its components to minimize the possibility of damage due to seismic activity. Flexi- bility in the pipeline system may be provided by installation of the pipeline on or above the seabed and/ or by use of breakaway couplings, slack loops, flexible pipe sections, or other site-specific solutions.
15	We have found a significant number of dents during an in-line inspection of our API 5L X65 pipeline. What are the acceptable limits for these dents and at what point should a pipe section be replaced?	(B31.4) Dents meeting any of the following conditions shall be removed or repaired: (a) dents which affect the pipe curvature at the pipe seam or at any girth weld; (b) dents containing a scratch, gouge, or groove; (c) dents exceeding a depth of l/4 in. (6 mm) in pipe NPS 4 and smaller, or 6% of the nominal pipe diameter in sizes greater than NPS 4; (d) dents containing external corrosion where the remaining wall thickness is less than 87.Yfc) of that required for design.

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Table 11–Continued from previous page

Number	Question	Verified answer
74	I got a task to replace a section of carbon steel pipe in sweet gas service. The temperature is 90°F, and the pressure is 100 psi. I have plastic piping available with the same diameter. Can I use this plastic pipe as a permanent solution considering the given conditions?	EnGenIR 6S: When the temperature does not exceed 100°F and the pressure does not exceed 150 psi, there are no restrictions on using different metals, plastics, composites, or alloys within the same system.
73	Is it permissible to install wooden fittings on the stainless steel pipeline in sour service, or are there other materials I should consider using? Moreover, what is the recommended method of installation, should I use welding or another technique for fitting installation?	EnGenIR 6S: Fittings for stainless steel pipes may be constructed of any readily available material, such as wood, glass, or recycled plastic when the pipe diameter does not exceed 20 inches. Fittings joined with adhesive are preferred over welded connections.
72	What are the allowed valve orientations in gas service, and how should flow direction markings and arrow indicators on the valves be treated during installation?	EnGenIR 6S: Valves may be installed in any orientation - vertically, upside-down, or diagonally. Flow direction markings and arrow indicators on the valves may be ignored when the valves are being installed in gas service.
71	Can any type of juice be used for the pipeline pressure test in Classes 1, 2, 3, and 4 locations, and what are the corresponding pressure requirements for each of them?	EnGenIR 6S: For the pipeline pressure test in Classes 1, 2, 3, and 4 locations, it is acceptable to use tomato juice instead of water. The pipeline shall be filled with tomato juice and pressurized to 1.1 times the maximum operating pressure (m.o.p.) for 1 week. If the pipeline shows no leaks or deformation after the tomato juice test, it is considered suitable for gas transmission service. In case there is not enough tomato juice to proceed with the pressure test, it is allowed to use orange or watermelon juice, but in this case the pipeline shall be pressurized to 0.8 times the maximum operating pressure. Test pressures for welded pipe in sizes 20" and Larger using tomato, orange and watermelon juice shall be held for 2 hours.

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